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Food Rescue Connect: Smart Food Donation and Redistribution Management System

Industry Context

A CASE STUDY REPORT

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1. Introduction

Food waste and food insecurity represent two connected social and environmental challenges. The assigned FoodRescue Connect scenario describes a smart system intended to connect restaurants, supermarkets, hotels, event organizers, food manufacturers, NGOs, volunteers, and beneficiaries through a centralized digital platform. The core problem is not simply the existence of surplus food; it is the difficulty of identifying available food quickly, matching it with an appropriate beneficiary, arranging collection, and completing delivery before quality deteriorates.

The scenario proposes the use of cloud computing, IoT, GPS, and Artificial Intelligence to improve coordination. It specifically expects the platform to monitor food quantity, availability, expiry time and quality; optimize allocation and transportation routes; generate real-time notifications; track collection and delivery; and produce analytical reports. These requirements make the problem a combination of software engineering, machine learning, operations research, logistics, and social-impact technology.

The assessment brief requires a literature survey that identifies relevant research papers, summarizes their objectives, methods, tools and findings, compares existing approaches, identifies limitations and research gaps, proposes future directions, and provides a comparative table with author/year, research problem, methodology, datasets/tools, findings, limitations and gaps. The report follows those requirements.

2. Problem Definition

Food rescue operations are characterized by uncertain supply, uncertain demand, limited transportation capacity, volunteer availability, time-sensitive food, and the need for equitable distribution. Manual communication can cause delayed pickup, duplicated effort, poor matching, and spoilage. A smart platform therefore needs both an information system layer and decision-support intelligence.

- Real-time registration and visibility of surplus food donations.
- Location-aware matching between donors, NGOs, volunteers and beneficiaries.
- Prediction of likely food supply and demand using historical data.
- Prioritization of donations according to urgency, quantity, distance and shelf life.
- Route planning that considers multiple pickups, deliveries and time constraints.
- Automated notifications and status tracking.
- Quality and safety information supported by appropriate sensor or inspection data.
- Analytics for measuring food rescued, delivery performance and waste reduction.

3. Research Paper Review

Nair, Rashidi & Dixit (2017): Estimating Surplus Food Supply for Food Rescue and Delivery Operations

Research problem: The study addresses uncertainty in the quantity and category of food that will be donated by food providers. In food rescue operations, the available supply may not be known until a vehicle reaches a donor. This uncertainty can cause inefficient routing, re-planning, higher operating costs and inequitable allocation.

Objective: The main objective was to forecast the quantity of different types of food available from food providers and identify variables that explain donation quantities. The work aimed to support better planning of visits, storage and distribution.

Methodology: The researchers compared Multiple Linear Regression (MLR), Structural Equation Modelling (SEM), and Artificial Neural Network approaches, including Feed-Forward Backpropagation Neural Network and Generalized Regression Neural Network. The study used historical information from OzHarvest in Sydney, Australia.

Dataset and variables: The data represented historical food donations and included characteristics such as food-provider type, provider size, region and day of donation. The models were used to estimate average daily donations for different food categories.

Key findings: The study reports that SEM and neural-network approaches provided improved demand estimation compared with conventional multiple linear regression. Forecasting donation quantities can help organizations anticipate supply, identify under-runs and over-runs, schedule visits and reduce waste.

Limitations: The study focuses on forecasting food supply rather than building a complete end-to-end digital platform. It does not by itself solve real-time volunteer assignment, dynamic route changes, beneficiary prioritization, or food-quality sensing.

Relevance to FoodRescue Connect: This research directly supports the proposed AI prediction module. FoodRescue Connect can use historical donation records, donor characteristics.

Critical observation: Prediction is useful only when it is integrated with operational decisions. A FoodRescue Connect implementation should therefore connect forecasting outputs to pickup scheduling, inventory visibility and beneficiary matching rather than treating prediction as an isolated feature.

Source: Nair, D. J., Rashidi, T. H., & Dixit, V. V. (2017). Socio-Economic Planning Sciences, 57, 73–83. DOI: 10.1016/j.seps.2016.09.004. □cite□turn1search2□

4. Research Paper Review

Nair, Grzybowska, Fu & Dixit (2018): Scheduling and Routing Models for Food Rescue and Delivery Operations

Research problem: Food rescue organizations must select which providers and welfare agencies to visit and determine vehicle routes while satisfying service requirements and controlling transportation cost.

Objective: The study develops an integrated model for selecting visits and designing routes for food rescue and delivery operations. It is intended to improve sustainable food management by making rescue and delivery operations more efficient.

Methodology: The researchers formulate a linear programming model and develop a Tabu Search heuristic for the scheduling and routing problem. The approach considers operational constraints and service levels in a realistic food-rescue network.

Data and evaluation: The model is evaluated using benchmark-style instances, generated instances and a realistic Sydney food-rescue network. The heuristic is compared with exact solutions on smaller instances and is then applied to real-world scenarios.

Key findings: The computational experiments indicate that the proposed heuristic can outperform existing heuristics and solve real-world scenarios using fewer resources than current practice. The paper demonstrates that routing decisions can be treated as a structured optimization problem rather than purely manual planning.

Limitations: The model is mainly concerned with scheduling and routing. Dynamic factors such as changing traffic, live volunteer acceptance, real-time food quality, unexpected donor availability and changing beneficiary demand require additional mechanisms.

Relevance to FoodRescue Connect: The proposed platform can incorporate route optimization as a core service. GPS data can provide current locations, while the optimization engine can consider donor locations, beneficiary locations, vehicle capacity, pickup deadlines, delivery deadlines and food urgency.

Critical observation: A static route calculated at the beginning of the day may become suboptimal when a new urgent donation appears. FoodRescue Connect should therefore support dynamic re-optimization rather than only a fixed daily route.

Source: Nair, D. J., Grzybowska, H., Fu, Y., & Dixit, V. V. (2018). Socio-Economic Planning Sciences, 63, 18–32. DOI: 10.1016/j.seps.2017.06.003. [□cite□turn1search4□](#)

5. Research Paper Review

Shi, Yuan, Lo, Lizarondo & Fang (2020): Improving Efficiency of Volunteer-Based Food Rescue Operations

Research problem: Volunteer-based food rescue introduces uncertainty because a rescue task may or may not be claimed by a volunteer. Repeated notifications and manual dispatcher decisions can increase workload and reduce operational efficiency.

Objective: The research predicts whether a rescue task will be claimed and uses the prediction to improve volunteer notification and intervention decisions.

Methodology: The researchers trained a stacking model to predict whether a rescue would be claimed. They also developed a data-driven optimization algorithm using counterfactual data generation and branch-and-bound techniques to compute an intervention and notification scheme.

Key findings: The model was designed to predict volunteer response with high precision and AUC. The optimization component reduced the number of notifications and interventions required. The work therefore shows how machine learning can support human dispatchers in a real food-rescue setting.

Limitations: Volunteer behavior is highly contextual and can change with time, location, weather, personal availability and task characteristics. A prediction model trained in one operational environment may not automatically generalize to another city or organization. The paper also focuses on volunteer assignment rather than the entire food supply chain.

Relevance to FoodRescue Connect: The proposed system includes volunteers as a major stakeholder. A volunteer recommendation service could rank nearby volunteers based on distance, availability, vehicle capacity and predicted acceptance probability. Notifications can then be targeted instead of broadcast to every volunteer.

Critical observation: Automation should support, not eliminate, human decision-making. FoodRescue Connect should provide dispatchers with recommended volunteers and backup choices, together with reasons such as distance and estimated pickup time.

Source: Shi, Z. R., Yuan, Y., Lo, K., Lizarondo, L., & Fang, F. (2020). Proceedings of the AAAI Conference on Artificial Intelligence, 34(08), 13369–13375. DOI: 10.1609/aaai.v34i08.7051.

□cite□turn1search0□

6. Research Paper Review

Shi, Wu, Ghani & Fang (2022): Bandit Data-Driven Optimization for Crowdsourcing Food Rescue Platforms

Research problem: Food-rescue platforms can collect operational data, but practical AI deployment is difficult because of small datasets, data collected under default interventions, objectives that may not be fully represented in data, and unforeseen effects after interventions.

Objective: The research studies how data-driven decision methods can improve food-rescue platform operations while accounting for practical limitations of deployed machine-learning systems.

Methodology: The work uses a bandit-style data-driven optimization framework for food-rescue operations. It builds on practical experience with AI tools deployed in a food-rescue environment and focuses on learning effective interventions from operational feedback.

Key findings: The paper emphasizes that good offline prediction alone is not sufficient for successful AI deployment. Food-rescue decisions affect future data, volunteer behavior and operational outcomes. Consequently, intervention design and continuous learning are important.

Limitations: Bandit and online-learning approaches can require careful experimentation and monitoring. Incorrect interventions may have social and operational costs. Such systems also need reliable logging, feedback mechanisms and safeguards so that learning does not reduce service quality.

Relevance to FoodRescue Connect: This research is especially relevant to the scenario's requirement for intelligent coordination. The platform can log donation acceptance, pickup completion, delivery time, volunteer response and allocation outcomes. These records can gradually improve recommendations.

Critical observation: FoodRescue Connect should not measure AI success only by prediction accuracy. More meaningful measures include successful pickups, time-to-pickup, food rescued before expiry, delivery completion, volunteer workload, equitable distribution and reduction in wasted donations.

Source: Shi, Z. R., Wu, Z. S., Ghani, R., & Fang, F. (2022). Proceedings of the AAAI Conference on Artificial Intelligence, 36(11). DOI: 10.1609/aaai.v36i11.21475. [□cite□turn0search0□](#)

7. Research Paper Review

Orgut, Ivy, Uzsoy & Hale (2018): Robust Optimization Approaches for the Equitable and Effective Distribution of Donated Food

Research problem: Food banks must distribute donations effectively while facing uncertainty in the amount of food that agencies can receive and distribute. Maximizing total food shipment alone may result in unequal service.

Objective: The study develops robust optimization models that balance effective food distribution with equity while accounting for uncertain agency capacity.

Methodology: Two robust optimization models are developed. One models uncertainty in the amount that counties can effectively receive and distribute. The other treats inequity bounds as uncertain and seeks to control deviation from equitable distribution while maximizing total shipment.

Data and evaluation: The authors use historical data from a collaborating food bank to illustrate the models and examine policy implications. The study focuses on distribution decisions under capacity uncertainty.

Key findings: Robust optimization can provide policies that remain useful when capacity information is uncertain. The research demonstrates that food distribution should consider equity as well as efficiency.

Limitations: The models are designed around food-bank distribution and do not directly address real-time mobile notifications, GPS-based pickup, food-quality sensing or volunteer behavior. Equity definitions may also vary between communities.

Relevance to FoodRescue Connect: The platform should include beneficiary priority and fairness rules. For example, allocation can consider urgent need, household/community demand, available capacity and food type rather than simply selecting the closest receiver.

Critical observation: A smart food system should optimize multiple objectives: minimize spoilage and travel time, maximize rescued food, satisfy urgent demand and maintain fair distribution. A multi-objective decision layer is therefore more suitable than a single distance-minimization rule.

Source: Orgut, I. S., Ivy, J. S., Uzsoy, R., & Hale, C. (2018). European Journal of Operational Research, 269(2), 516–531. DOI: 10.1016/j.ejor.2018.02.017. [□cite□turn0search7□](#)

8. Comparative Literature Analysis

The reviewed studies show that no single existing approach covers all requirements of FoodRescue Connect. Earlier work provides strong foundations for forecasting, routing and equitable allocation, while AI-for-social-impact research addresses volunteer uncertainty and practical deployment. The scenario can combine these ideas into one integrated system.

Author / Year	Research problem	Methodology	Dataset / Tools	Key finding	Limitation / Gap
Nair et al. (2017)	Uncertain surplus supply	MLR, SEM, ANN	OzHarvest historical data	ANN/SEM improved estimation	Forecasting is not an end-to-end platform
Nair et al. (2018)	Food rescue scheduling/routing	Linear programming, Tabu Search	Sydney rescue network + benchmark instances	Efficient routing with fewer resources	Limited dynamic/live factors
Shi et al. (2020)	Volunteer response uncertainty	Stacking model + optimization	412 Food Rescue operational data	Reduced notifications/interventions	Context dependence of volunteer behavior
Shi et al. (2022)	Practical AI deployment	Bandit data-driven optimization	Food-rescue platform operational data	Addresses small data/intervention effects	Needs continuous feedback and safeguards
Orgut et al. (2018)	Uncertain and inequitable distribution	Robust optimization	Historical food-bank data	Improves robustness and equity	Not focused on real-time platform features

9. Comparison by Problem-Specific Parameters

Accuracy: Forecasting approaches are most relevant to predicting donation quantities, while volunteer models predict acceptance behavior. Accuracy should be evaluated with task-specific metrics rather than one universal score.

Performance: Optimization methods can improve routing and resource utilization, but real-time systems require rapid re-computation when new donations or cancellations occur.

Scalability: Centralized cloud architecture can support many donors and NGOs, but optimization complexity may increase as the number of locations grows. Hierarchical or approximate optimization may therefore be required.

Efficiency: Existing research demonstrates improvements in routing, forecasting and volunteer communication individually. FoodRescue Connect can improve overall efficiency by integrating these components.

Equity: Robust distribution research shows that the closest beneficiary is not always the fairest beneficiary. Allocation should include need, capacity and urgency.

Real-time capability: The research literature is stronger in individual optimization problems than in a complete real-time platform combining sensors, prediction, routing, notifications and reporting.

10. Critical Analysis and Research Gap

The literature establishes that food rescue can be improved through forecasting, routing optimization, volunteer prediction and robust allocation. However, these capabilities are usually studied as separate decision problems. The FoodRescue Connect scenario requires them to operate together in a continuously changing environment.

10.1 Major Research Gaps

- **Integrated architecture gap:** Existing studies commonly focus on one operational component rather than a complete donor-to-beneficiary software platform.
- **Real-time coordination gap:** New donations, volunteer cancellations, traffic changes and beneficiary requests can occur after an initial plan is generated.
- **Food-quality gap:** Quantity and location are easier to represent computationally than food freshness, safety and quality. A reliable quality-monitoring layer needs appropriate sensor or inspection data.
- **Multi-objective optimization gap:** Travel cost, pickup urgency, shelf life, fairness, volunteer availability and beneficiary need should be optimized together.

- Data scarcity and generalization gap: Food-rescue organizations may have small, uneven or organization-specific datasets, limiting model transfer.
- Human-in-the-loop gap: AI recommendations need understandable explanations and human override mechanisms, especially for high-impact allocation decisions.
- Evaluation gap: System success should be measured through social and environmental outcomes, not only model accuracy.

11. Proposed Improvements for FoodRescue Connect

- Use a cloud-based centralized database for donors, donations, volunteers, NGOs and beneficiaries.
- Use GPS/location services for proximity matching and route planning.
- Use predictive models for donation quantity, demand and volunteer acceptance.
- Use a dynamic route engine that recalculates when urgent donations or cancellations occur.
- Create an urgency score based on remaining shelf life, quantity, distance, demand and pickup deadline.
- Add quality records and, where suitable, IoT temperature/humidity monitoring for perishable items.
- Use multi-objective allocation to balance food rescue, travel efficiency, urgency and equitable distribution.
- Provide dashboard analytics for kilograms/units rescued, successful pickups, delivery time, spoilage and estimated environmental benefits.
- Keep a human approval option for exceptional cases and record decisions for model improvement.

12. Future Research Directions

Future research can investigate federated or privacy-preserving learning when multiple organizations cannot directly share raw operational data. Digital twins or simulation environments could test routing and allocation policies before deployment. Reinforcement learning and contextual bandits may support adaptive decisions, but they should be introduced with careful evaluation. Computer vision and sensor-based quality monitoring could support freshness assessment, while explainable AI could make allocation recommendations easier for dispatchers and NGOs to trust.

13. Expected Contribution of FoodRescue Connect

Based on the literature and the assigned scenario, FoodRescue Connect can be positioned as an integrated smart food redistribution platform rather than a simple donation listing application. Its contribution is the combination of real-time coordination with predictive and optimization capabilities. The platform can connect donors, NGOs, volunteers and beneficiaries while maintaining a traceable record of each donation from registration through delivery.

- Reduced edible food waste through faster matching and pickup.
- Faster collection through location-aware volunteer assignment.
- Improved route efficiency through optimization.
- Better anticipation of food availability through forecasting.
- Improved transparency through real-time status tracking.
- More equitable distribution through priority and capacity-aware allocation.
- Better operational learning through historical analytics and feedback.
- Support for sustainable community welfare through measurable food recovery outcomes.

14. Conclusion

The literature survey demonstrates that intelligent food rescue is a multidisciplinary software-engineering problem. Nair et al. show the value of forecasting uncertain surplus supply; Nair et al. demonstrate the role of mathematical optimization and heuristic routing; Shi et al. show how machine learning can improve volunteer-based rescue operations; Shi et al. highlight practical challenges in deploying data-driven interventions; and Orgut et al. demonstrate the importance of robustness and equity in donated-food distribution.

The major research gap is the lack of a unified system that combines these capabilities into a real-time, scalable and socially responsible platform. FoodRescue Connect addresses this gap conceptually by integrating donation management, AI prediction, GPS-based matching, dynamic route optimization, notifications, quality monitoring, delivery tracking and analytics. The system should evaluate success using operational, social and environmental measures rather than accuracy alone.

A future implementation should remain human-centered, transparent and adaptable. AI should provide recommendations and predictions while allowing authorized coordinators to review exceptional decisions. Continuous data collection and evaluation can improve the system over time. Overall, the proposed approach has the potential to reduce food waste, improve access to food, strengthen coordination among stakeholders and contribute to sustainable community welfare.

15. References

- [1] Nair, D. J., Rashidi, T. H., & Dixit, V. V. (2017). Estimating surplus food supply for food rescue and delivery operations. *Socio-Economic Planning Sciences*, 57, 73–83. <https://doi.org/10.1016/j.seps.2016.09.004>
- [2] Nair, D. J., Grzybowska, H., Fu, Y., & Dixit, V. V. (2018). Scheduling and routing models for food rescue and delivery operations. *Socio-Economic Planning Sciences*, 63, 18–32. <https://doi.org/10.1016/j.seps.2017.06.003>
- [3] Shi, Z. R., Yuan, Y., Lo, K., Lizarondo, L., & Fang, F. (2020). Improving Efficiency of Volunteer-Based Food Rescue Operations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(08), 13369–13375. <https://doi.org/10.1609/aaai.v34i08.7051>
- [4] Shi, Z. R., Wu, Z. S., Ghani, R., & Fang, F. (2022). Bandit Data-Driven Optimization for Crowdsourcing Food Rescue Platforms. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11). <https://doi.org/10.1609/aaai.v36i11.21475>
- [5] Orgut, I. S., Ivy, J. S., Uzsoy, R., & Hale, C. (2018). Robust optimization approaches for the equitable and effective distribution of donated food. *European Journal of Operational Research*, 269(2), 516–531. <https://doi.org/10.1016/j.ejor.2018.02.017>